Apparent Performance under Selective Environmental Risk: Strategic Acceptance and Reputation under a Zero-Effect Rainfall Null

Abstract

When providers choose which cases to accept, raw outcome rates combine baseline risk, selective exposure, and any causal provider effect—and the untreated counterfactual is usually unobserved. We impose an exact zero-effect null, $R(1)=R(0)$, so that every mechanism other than efficacy is isolated. Using daily ERA5 precipitation at eight pre-specified climate stress-test locations, we estimate prospective rainfall hazards and construct fictional observed-weather stress tests; a separate calibrated Monte Carlo model adds drought-driven demand, selective acceptance, scenario willingness to pay (WTP), Bayesian reputation, and selective memory. Among 2595 fictional events, seven-day apparent success is 0.593 under random drought timing and 0.853 under high prospective hazard. Selective acceptance changes accepted-event composition and raises mean simulated apparent success from 0.493 to 0.676 without changing rainfall. Reputation models generate mean scenario prices of 3.568 and 4.610 above a fixed 3.000 baseline, yet mean contemporaneous price-success differences are -2.447 and -3.037; a cluster-robust scenario-WTP interval includes zero and an exploratory permutation diagnostic is negative. WTP and prices are calibrated or simulated, not historical estimates. Raw observed performance can therefore reward favorable risk exposure and selective acceptance rather than causal performance; past attributed outcomes form a reputation stock, while current performance remains a hazard-driven flow.

Keywords: performance evaluation; risk adjustment; selective exposure; attribution; reputation

1. INTRODUCTION

Performance evaluation is difficult when baseline risk varies. A provider whose outcomes are scored on raw success can look effective by taking favorable cases; an evaluator who cannot observe the untreated counterfactual cannot separate baseline hazard, selection into exposure, and causal performance. In forecasting, goodness is assessed relative to the difficulty of the prediction task (Murphy, 1993); in health outcomes research, risk adjustment attempts the same correction for provider comparisons (Shahian & Normand, 2008). Wherever agents choose their exposure, selection into observation confounds raw performance measures (Heckman, 1979), and public scorecards can reward case selection rather than improvement (Dranove, Kessler, McClellan, & Satterthwaite, 2003; Werner & Asch, 2005).

Hidden counterfactuals make the problem worse. Interventions are often initiated after unusually adverse conditions, creating regression-to-the-mean risk (Barnett et al., 2004). Observers use heuristics under uncertainty (Tversky & Kahneman, 1974), intermittent outcomes sustain beliefs (Skinner, 1948), and social learning can propagate an attribution once it forms (Bikhchandani, Hirshleifer, & Welch, 1992). Reputation can then accumulate under imperfect information (Kreps & Wilson, 1982).

Rainmaking offers a deliberately sharp known-ground-truth stress test: the true causal effect on rainfall is exactly zero. Espín-Sánchez, Gil-Guirado, and Ryan (2026) develop an instrumental-belief account in which leaders cannot produce rain, drought timing and changing rainfall hazards make prayer persuasive, untreated outcomes are hidden, and persuasive practices can persist. Their evidence includes long-run Murcia records and a large cross-cultural classification, and their theory discusses tangible benefits, support costs, and a strategic-timing alternative; we

concede those contributions and claim no novelty there. Historical evidence places rain-prayer institutions within broader problems of scarcity, resilience, and social stability (Espín-Sánchez & Gil-Guirado, 2022). Comparative ethnography describes rainmaking responses to rainfall scarcity or unreliability (Akong'a, 1987), and other accounts document specialized roles and stated payments (Semenya, 2013). These sources support institutional plausibility, not a universal law that providers screened requests or raised fees with drought; prices and acceptance strategies below are calibrated scenarios rather than reconstructed history.

Our contribution is to the performance-evaluation problem. We separate requests from accepted cases, allow multiple providers that differ in selectivity, let past attributed outcomes create provider-specific reputation stocks, and let those stocks interact with current hazard in simulated willingness to pay (WTP) and price. Because efficacy is fixed at exactly zero, observed differences in raw performance arise only from baseline hazard, selection, attribution, and learning. The principal result is unfavorable to a simple performance-pricing story: reputation-mediated price premia coexist with zero or negative mean contemporaneous price-success differences.

Four distinctions organize the analysis. First, causal efficacy is a potential-outcome contrast, whereas apparent success is rainfall inside an attribution window. Second, baseline rainfall hazard is a risk state, not evidence of treatment response. Third, mean scenario price, provider-price heterogeneity, a reputation-mediated premium, and the contemporaneous price-success association are different objects. Fourth, scenario WTP, a theoretical revenue-maximizing price, expenditure transfers, resource costs, and non-monetary value are not interchangeable. A separate social-utility term is retained so that null meteorological efficacy cannot be misread as

null social value; plural environmental values are documented elsewhere (Chan et al., 2012; Kenter et al., 2015; Pascual et al., 2017; Himes & Muraca, 2018) and are not estimated here. We examine eight declared analytic propositions: natural rainfall can generate apparent success; drought-driven demand increases intervention frequency; drought can increase scenario WTP; selective acceptance increases apparent success; higher prices are positively associated with apparent success; learning creates persistent provider differentiation; selective memory raises price premia; and zero meteorological efficacy does not logically imply zero social value.

2. CONCEPTUAL FRAMEWORK

Figure 1 represents the data-generating system. Baseline environmental hazard enters demand, provider acceptance, valuation, attribution, and reputation, while the pseudo-intervention never changes rainfall. The dashed separation is the central exclusion: there is no causal arrow from the pseudo-event to rainfall.

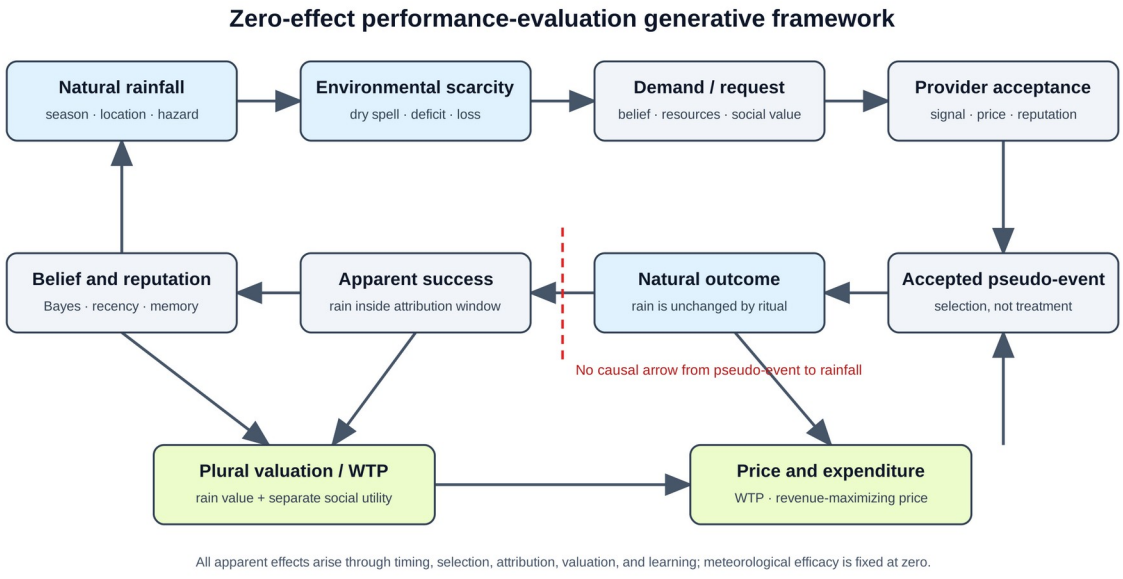


Figure 1. Conceptual framework. Baseline environmental hazard enters demand, provider acceptance, valuation, attribution, and reputation, while the pseudo-intervention never changes rainfall.

2.1. Exact zero-effect null

For location r, day t, and pseudo-intervention indicator A_{rt}, let $R_{rt}(a)$ denote potential rainfall under $a \in \{0,1\}$. The simulation enforces

$$R_{rt}(1) = R_{rt}(0) = R_{rt}.$$

No request, acceptance, provider, price, belief, or reputation variable changes this array.

Apparent success for window w and rainfall threshold c is

$$S_{rt}(w,c) = 1\{max_{j=1,\ldots,w} R_{r,t+j} \geq c\}.$$

The event day is excluded. This definition captures a plausible retrospective attribution rule but is not an estimator of a treatment effect.

2.2. Baseline hazard, demand, and selective exposure

Let $H_{rt}(w,c)$ be the prospective probability of apparent success conditional on location, calendar month, and the preceding dry spell—the risk-conditioning variable for this study. We estimate this hazard from a training period rather than assume it rises with drought duration.

Provider k observes a signal of H_{rt} and accepts a request with a probability increasing in that signal. Selective acceptance can therefore raise the success rate among accepted events even though it cannot change rainfall.

Drought affects demand separately from selection. A community is more likely to request action when the preceding dry spell is long or the rolling rainfall deficit is large. This changes the

timing and frequency of observed events. It does not imply that rain is more likely merely because the dry spell is longer.

2.3. Reputation stock, current flow, and price

Scenario WTP combines an instrumental component related to expected avoided drought loss with a separate social-utility component. This construction is motivated by empirical evidence that people value drought-risk adaptation and supply security (Schrieks et al., 2025; Pattanayak & Kramer, 2001; Hensher et al., 2006), but its numerical scale is not estimated from historical rainmaking payments. Payments are transfers from communities to providers; provider costs are resource costs; profit is revenue minus those costs. Because the pseudo-intervention creates no rainfall, its actuarially fair causal rainfall price is zero. The theoretical revenue-maximizing price is also distinct from the implemented quote, which is a fixed share of calibrated WTP.

Observers update provider beliefs only after accepted events. A Bayesian observer applies a Beta-Binomial update. A selective-memory observer gives greater weight to apparent successes than failures and discounts older observations. In M4–M5, the reputation stock ρ_{jt} is determined by outcomes before t, while current hazard H_{rt} enters the simulated quote:

$$WTP_{jrt} = B(D_{rt})[U^S + V^R(D_{rt}) max(0, \rho_{jt} - H_{rt})], p_{jrt} = 0.8\, WTP_{jrt}.$$

Current apparent success remains natural rainfall after the event. Past attributed outcomes can therefore create a reputation-mediated price premium and provider-price heterogeneity without making current price positively associated with current success. Informational cascades (Bikhchandani et al., 1992) and reputation under imperfect information (Kreps & Wilson, 1982) provide familiar economic analogues, but the mechanisms remain calibrated and stylized.

3. METHODS

3.1. Design and rainfall data

The study uses daily precipitation from 1 January 1991 through 31 December 2020 at eight pre-specified point locations spanning contrasting rainfall regimes. The points are mechanism stress tests, not identified ritual communities or a representative global sample. Table 1 reports coordinates, regimes, observations, and missingness. The analyzed panel contains 87664 region-days.

Table 1. Study locations and rainfall observations.

Location	Latitude	Longitude	Rainfall regime	Region-days	Missing precipitation
Tucson, United States	32.223	-110.975	hot semi-arid / monsoonal	10958	0
Cape Town, South Africa	-33.925	18.424	Mediterranean winter-rainfall	10958	0
London, United Kingdom	51.507	-0.128	temperate maritime	10958	0
Mumbai, India	19.076	72.878	tropical monsoon	10958	0
Nairobi, Kenya	-1.292	36.822	subtropical highland / bimodal rainfall	10958	0
Darwin, Australia	-12.463	130.846	tropical savanna / monsoon	10958	0
Tokyo, Japan	35.676	139.650	humid subtropical	10958	0
Brasília, Brazil	-15.794	-47.883	tropical savanna	10958	0

Rainfall was acquired through the Open-Meteo historical archive with the ERA5 model. ERA5 is a global atmospheric reanalysis (Hersbach et al., 2020). Raw JSON responses are preserved with URLs, UTC retrieval times, byte sizes, and SHA-256 checksums. An initial ERA5-Land request

returned complete date arrays but all precipitation values were null; those snapshots are retained as invalid audit material and excluded.

3.2. Prospective baseline hazard

The years 1991–2010 form the training period and 2011–2020 the evaluation period. The primary rain-day threshold is 1 mm, with 0.1 and 5 mm sensitivity thresholds. Apparent success windows are 1, 3, 7, and 14 days. Preceding dry-spell duration and 30-day rainfall totals use only days before a candidate event. Candidate dates in both periods are truncated 14 days before the period boundary so every outcome window has complete follow-up. Training hazards condition on location, month, and dry-spell bins. This flexible stratification is used instead of a monotonic duration assumption. Cells with fewer than 20 training days use the corresponding location-month mean. The high-hazard threshold is the training-cell 75th percentile within location and never uses evaluation-period rainfall. Duration-based hazards are a descriptive survival object rather than a treatment response (Cox, 1972).

3.3. Known-zero-effect stress tests

Five fictional event rules are constructed from evaluation-period days without using event-day or future rainfall:

1. a dry-spell threshold;
2. a 30-day rainfall-deficit threshold;
3. a random drought day;
4. a high prospective-hazard threshold; and
5. a season-matched placebo.

Event-day rainfall is unavailable to every rule. Outcomes are always future observed rainfall, never generated or modified by an event. Rule-specific apparent success is reported across all thresholds and windows.

3.4. Scenario WTP and descriptive association

Scenario WTP increases with drought loss, social utility, and perceived probability of success. Rainfall and pseudo-event dates in this component are observed; WTP is constructed from calibrated quantities. We estimate a naive logit of seven-day apparent success on scenario WTP and a risk-conditioned model including prospective hazard, drought severity, location, calendar month, and timing-rule effects. Standard errors are two-way clustered by region-year and region-rule, allowing arbitrary dependence within the weather-year and timing-rule series. Because rules can select the same or nearby dates, attribution windows can overlap, WTP is constructed from calibrated quantities, and the event is fictional, these regressions are exploratory model-consistent associations rather than estimates of demand, historical prices, or causal efficacy.

3.5. Staged Monte Carlo model comparison

The simulation samples one observed evaluation-year rainfall sequence per replication and uses the same sampled region-year across M0–M5. M0–M1 independently permute outcomes within that year to remove calendar structure; M2–M5 retain the observed local and seasonal ordering. The sequence is a mechanism decomposition or sequential generative enrichment, not a clean one-factor causal decomposition: restoring ordering between M1 and M2 changes multiple properties of the weather process. Each model has 10,000 replications and four providers:

- **M0 Random timing:** exogenous requests and shuffled outcomes.
- **M1 Drought demand:** requests respond to drought, with shuffled outcomes.
- **M2 Realistic hazard:** drought demand operates on local seasonal rainfall.

- **M3 Selective acceptance:** providers condition acceptance on prospective hazard.
- **M4 Bayesian reputation:** outcomes update provider beliefs and belief-mediated WTP.
- **M5 Selective/recency memory:** successes and failures receive unequal, discounted weights.

Primary outputs are event frequency, apparent success, mean scenario price, provider-mean-price dispersion, event-level price–success difference, reputation dispersion, provider revenue, community expenditure, provider resource cost, provider profit, community material-plus-social utility, total social surplus, and provider-revenue dispersion. Event-level price–success differences are defined only for replications containing both successes and failures; event-based quantities are undefined when no event is accepted, and summaries omit only those undefined cells.

3.6. Robustness and falsification

Robustness analysis varies rainfall thresholds and attribution windows, provider selection slopes, Bayesian prior means, and memory weights. A 1,000-permutation falsification demeans outcomes and WTP within region–calendar-month–timing-rule strata, adjusts for hazard and drought severity, and shuffles WTP within the same strata. Exchangeability is assumed only within those strata. Monte Carlo precision is evaluated at 500, 1,000, 2,500, 5,000, and 10,000 replications. All propositions retain unfavorable or mixed results; no parameter was retuned to manufacture the anticipated sign. P-values are descriptive and exploratory: no multiplicity adjustment is used, and classifications rely on the full pattern across observed-weather, falsification, and simulation results rather than a 0.05 threshold. Detailed models, sensitivities, convergence, falsification, and accounting results are reported in the Supplementary material (Tables S1–S6).

4. RESULTS

4.1. Baseline hazard is heterogeneous and not monotone in dry-spell duration

Figure 2 shows substantial seasonal and geographic variation in seven-day rainfall hazards. The ordering of dry-spell bins is not uniformly increasing: a longer preceding dry spell can coincide with higher, lower, or similar prospective rainfall probability depending on location and month. The empirical hazard therefore rejects the shortcut that rain must become mechanically more likely as a dry spell lengthens.

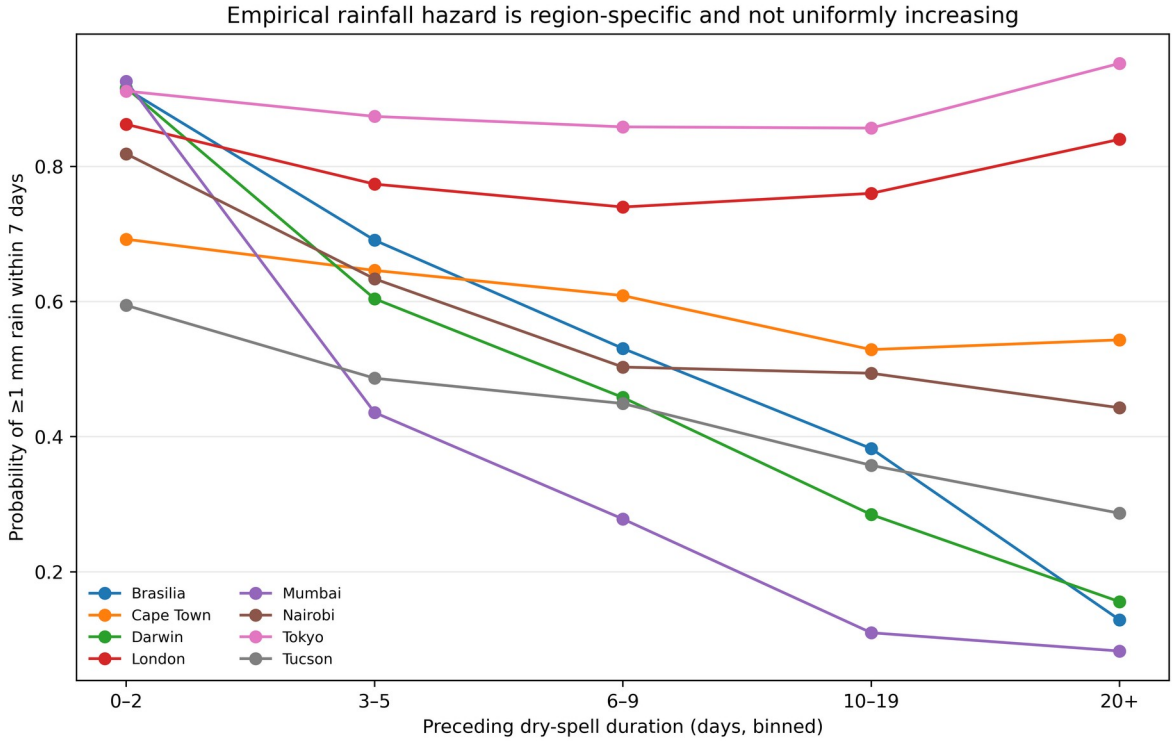


Figure 2. Estimated seven-day rainfall hazards by preceding dry-spell duration and location in the 1991–2010 training period. The eight locations are pre-specified mechanism stress tests, not a representative global sample.

4.2. Raw apparent performance varies under exact zero efficacy

The five event rules produce 2595 fictional events. Table 2 reports counts and success across attribution windows. At the primary seven-day, 1-mm outcome, apparent success is 0.823 for the rainfall-deficit rule, 0.601 for the dry-spell rule, 0.853 for high prospective hazard, 0.593 for a random drought day, and 0.600 for the season-matched placebo.

Table 2. Observed-weather zero-effect stress-test results.

Selection rule	Events	Apparent success 1d	Apparent success 3d	Apparent success 7d	Apparent success 14d	Mean scenario WTP
30-day deficit threshold	294	0.374	0.592	0.823	0.922	6.276
dry-spell threshold	727	0.209	0.402	0.601	0.765	3.267
high prospective hazard	395	0.461	0.671	0.853	0.947	6.815
random drought day	509	0.191	0.381	0.593	0.745	3.073
season-matched placebo	670	0.245	0.407	0.600	0.793	3.550

Figure 3 emphasizes the mechanism: prospective-hazard selection has the highest apparent success, whereas drought timing alone does not guarantee a favorable counterfactual comparison. Every bar is generated by natural rainfall under the exact zero-effect null.

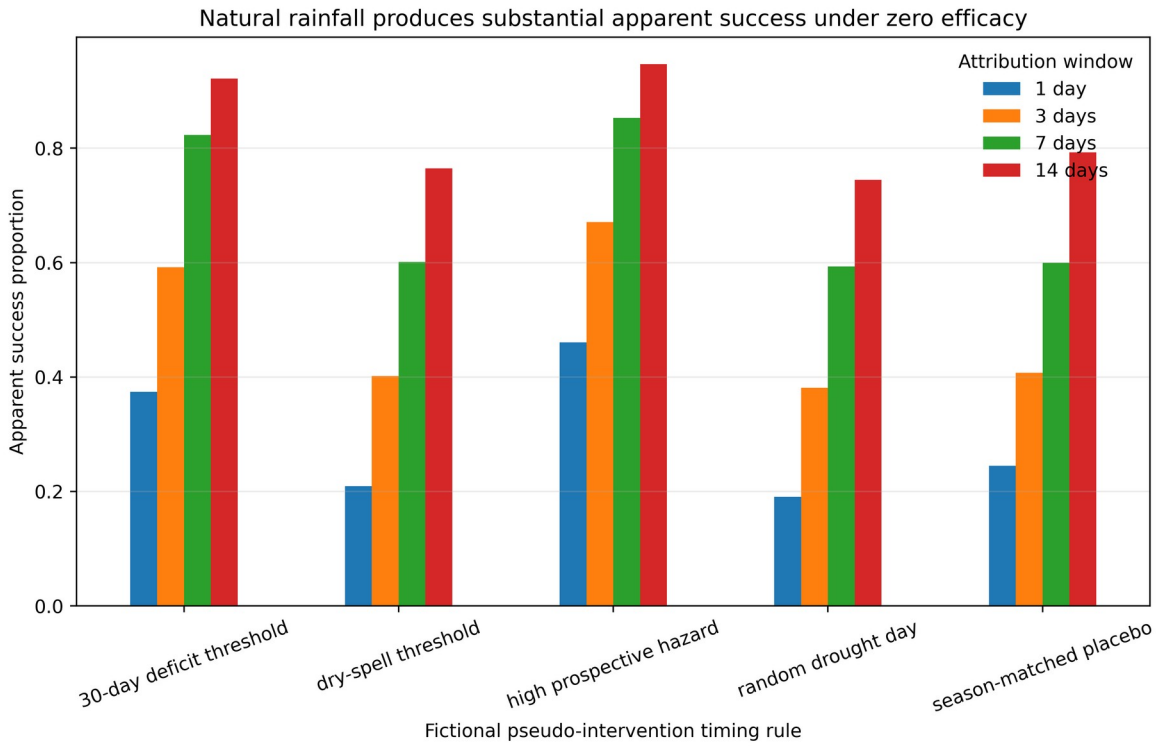


Figure 3. Natural rainfall following fictional known-zero-effect stress tests in 2011–2020, shown as apparent success by prospective event-selection rule.

4.3. Selective acceptance changes observed risk composition

Table 3 reports the staged model comparison. Moving from M0 to M1 increases mean event frequency from 24.36 to 68.95 while apparent success remains near 0.635 and 0.634. Restoring realistic local and seasonal rainfall in M2 changes apparent success to 0.493. Selective acceptance in M3 reduces the number of accepted events to 16.92 and raises apparent success to 0.676, with rainfall still unchanged. This M2–M3 contrast shows how the acceptance rule changes accepted-event composition; it is not an estimate of historical provider behavior.

Table 3. Monte Carlo model decomposition.

Model	apparent success	events	mean price	price success difference	provider price dispersion	reputation dispersion
M0 Random timing	0.635	24.362	3.000	0.000	0.000	0.000
M1 Drought demand	0.634	68.952	3.000	0.000	0.000	0.000
M2 Realistic hazard	0.493	68.978	3.000	0.000	0.000	0.000
M3 Selective acceptance	0.676	16.917	3.000	0.000	0.000	0.000
M4 Bayesian reputation	0.674	16.946	3.568	-2.447	0.887	0.117
M5 Selective/recency memory	0.673	16.882	4.610	-3.037	1.373	0.138

4.4. Risk conditioning changes the naive WTP-success association

The naive scenario-WTP coefficient is 0.124 (95% model-based interval 0.086–0.162). Conditioning on prospective hazard and drought covariates reduces it to 0.036 (-0.028–0.100). Two-way clustering gives a 95% interval of -0.039–0.111 across 80 clusters by region-year and 40 by region-rule ($n = 2595$).

The adjusted linear-probability coefficient is -0.0319, compared with a permutation mean of -0.0002 and two-sided $p = 0.002$. The permutation diagnostic has a negative within-stratum association rather than the declared positive one. Because it assumes exchangeability within the defined strata, uses a constructed scenario-WTP regressor, and retains overlapping outcome windows, it is exploratory rather than confirmatory historical evidence. Together with the negative mean M4–M5 event-level price–success differences below, it does not support the general positive price-success proposition.

4.5. Reputation and price persist as stocks while current performance remains hazard-driven

M0–M3 use a fixed scenario price of 3.000. Bayesian reputation in M4 generates mean scenario price 3.568, a mean premium of 0.568, reputation dispersion 0.117, and provider-mean-price dispersion 0.887. Selective/recency memory in M5 produces 4.610, 1.610, 0.138, and 1.373, respectively. These are calibrated simulation quantities, not historical currencies or estimates. Contrary to a simple performance-pricing story, the mean event-level price(success)-price(failure) difference is -2.447 in M4 and -3.037 in M5. Thus mean price relative to baseline, provider price heterogeneity, and a reputation-mediated premium can all be positive while the contemporaneous price-success difference is negative. Past attributed outcomes enter the reputation stock, whereas current apparent success remains a flow from current natural hazard. Figure 4 visualizes these margins. Drought demand expands event frequency; provider selection changes accepted-event composition and current success; learning creates reputation and provider-price dispersion; and price premia coexist with negative mean contemporaneous price-success differences.

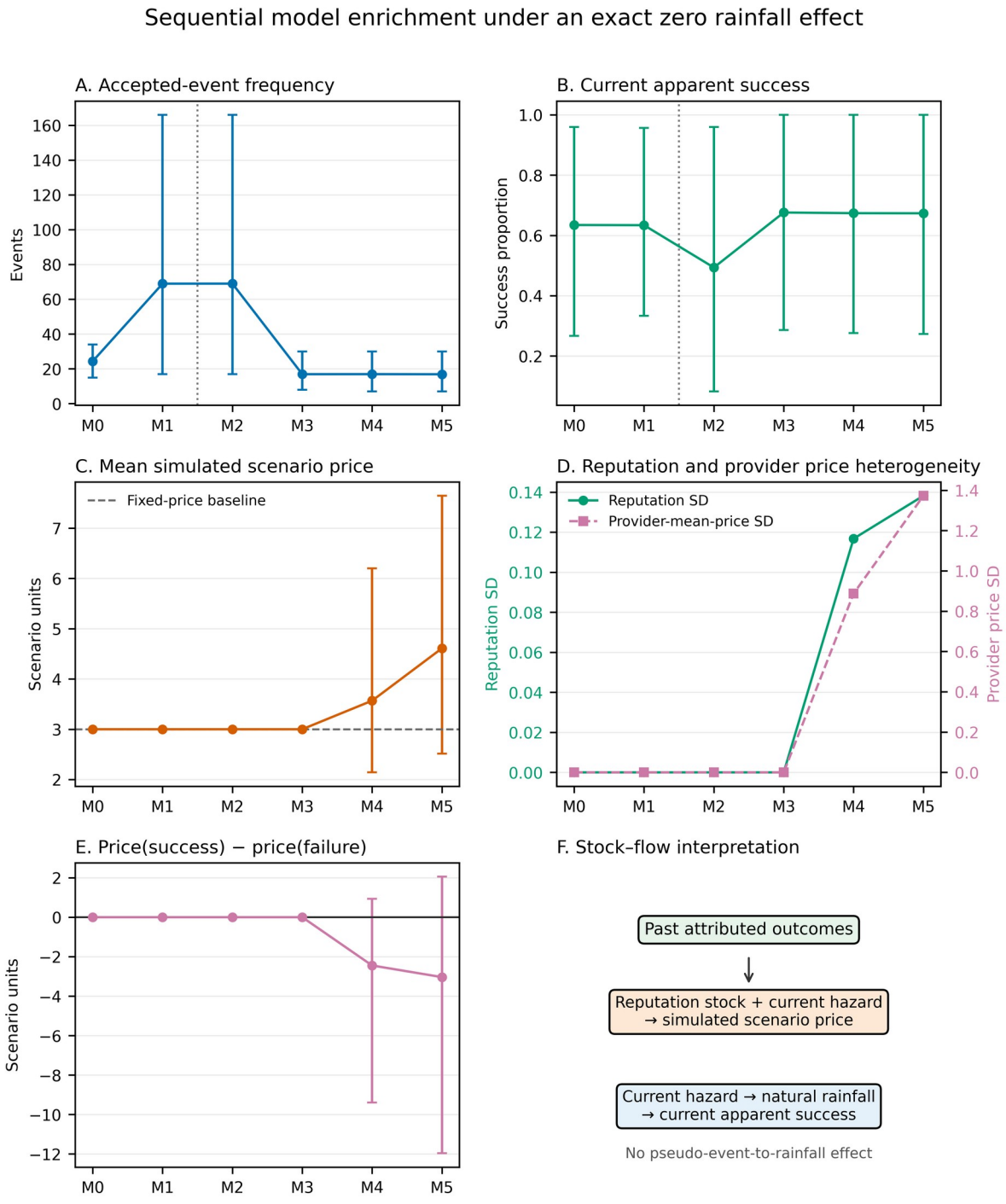


Figure 4. Sequential Monte Carlo model enrichment under M0–M5. M0–M1 shuffle rainfall outcomes within sampled years; M2–M5 retain observed local and seasonal ordering. Selective acceptance changes accepted-event composition and current apparent

success. In M4–M5, past attributed outcomes accumulate in reputation stocks that create price premia and provider-price heterogeneity, while current success remains a natural-rainfall flow governed by current hazard. Mean price(success) minus price(failure) is negative in M4–M5. Error bars are 2.5th and 97.5th Monte Carlo percentiles, not confidence intervals for historical parameters.

4.6. Findings are stable across thresholds, parameters, and replication counts

Figure 5A shows that apparent success increases with the attribution window and falls with the rainfall threshold, as expected from the definition. Figure 5B shows monotonic increases in M3 apparent success as the selection slope rises from 2 to 8. M4 prices increase with the Bayesian prior mean, and M5 prices increase with stronger selective memory. Figure 5C shows falling Monte Carlo standard errors; at 10,000 replications, apparent-success standard errors are at most 0.0028 among the six main models.

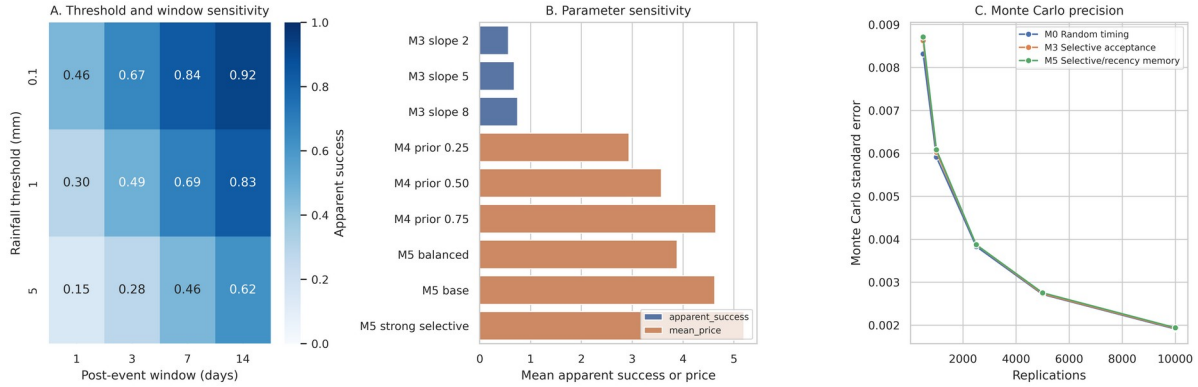


Figure 5. Robustness. Apparent success across rainfall thresholds and windows, selected parameter sensitivity, and Monte Carlo precision.

Table 4 classifies the analytic propositions by evidence domain. Natural-rainfall apparent success is demonstrated with observed weather, whereas selective acceptance is demonstrated in the model using observed weather. Demand, WTP, learning, and memory results are model- or calibration-contingent; the price–success proposition is not supported; and the separation of meteorological from social value is a logical distinction rather than an empirical finding.

Table 4. Analytic proposition classification.

Hypothesis	Classification	Basis
Natural rainfall can generate substantial apparent success under zero efficacy	demonstrated with observed weather	Observed-weather pseudo-events and all simulation models
Drought-driven demand increases intervention frequency	demonstrated in model	M1 versus M0 under the specified demand function
Drought can increase scenario WTP	demonstrated in calibrated scenario	Calibrated rainfall-loss term raises WTP over intermediate drought severity but is offset at extremes by ability to pay
Strategic provider selection increases apparent success	demonstrated in model using observed weather	M3 versus M2 and selection-slope sensitivity
Higher prices are positively associated with apparent success	not supported	Two-way clustered logit interval includes zero; within-stratum permutation estimate and M4-M5 differences are negative
Reputation learning creates persistent provider differentiation	demonstrated in model	M4-M5 reputation dispersion under specified update rules
Selective and recency-weighted memory raises price premia	demonstrated in model	M5 versus M4 and memory sensitivity
Zero meteorological efficacy implies zero social value	logical distinction; not empirically tested	Positive social utility is modeled separately from causal rainfall benefit

5. DISCUSSION

5.1. Selective exposure can masquerade as performance

The QJE comparator already establishes that endogenous prayer timing, changing rainfall hazards, hidden counterfactuals, and strategic leader timing can make an ineffective practice persuasive (Espín-Sánchez et al., 2026). Our narrower extension separates requests from acceptance and adds multiple provider identities, provider-specific reputation stocks, simulated provider prices, and explicit accounting.

Scarcity changes who requests action and when; the calibrated provider rule changes which requests become accepted events; attribution maps natural rainfall into apparent success; and learning maps past outcomes into reputation. Demand can increase without success changing. Selection can increase current success without treatment. Reputation can persist without provider-specific efficacy. Price premia can arise without positive contemporaneous price-success associations.

5.2. Raw performance versus risk-adjusted performance

When baseline risk is heterogeneous and exposure is selective, raw outcome rates answer the wrong question. They combine the baseline risk of accepted cases, the selection rule, and any causal effect; this benchmark fixes the last term at zero, so the apparent provider differences arise entirely from the first two plus attribution and learning. Risk adjustment exists precisely because evaluators who ignore baseline differences reward favorable case mix rather than performance (Shahian & Normand, 2008), and public scorecards have induced selection rather than improvement in other settings (Dranove et al., 2003; Werner & Asch, 2005). Prospective

hazard estimates like ours supply the conditioning variable such an adjustment would need—but they condition on a limited state vector and do not identify causal effects by themselves.

5.3. Reputation is a stock; current outcome is a risk-conditioned flow

The unfavorable price result is substantively informative. In the observed-weather scenario, the adjusted logit point estimate is positive, but the two-way clustered interval includes zero; the exploratory within-stratum permutation diagnostic is negative. In M4 and M5, mean event-level price-success differences are also negative, although their Monte Carlo percentile intervals include positive values. Reputation is a stock formed from earlier attributed outcomes, whereas current apparent success is a flow generated by current hazard. A high-reputation provider can charge more when the reputation-hazard gap is large and still encounter unfavorable natural rainfall.

This is a boundary condition, not a defect to be removed by retuning. Mean prices, provider-price heterogeneity, and reputation premia can reflect accumulated belief without identifying current causal performance. The findings motivate comparing provider outcomes with prospective risk-adjusted benchmarks rather than raw success, and separating reputation from current causal performance rather than treating it as its measure.

5.4. Decision and institutional implications

For provider scorecards and public performance reporting, the benchmark warns that raw outcome comparisons can reward selective acceptance and favorable exposure; risk-adjustment and disclosure of baseline hazard—and of rejected cases—may serve as candidate safeguards (Dranove et al., 2003; Werner & Asch, 2005; Shahian & Normand, 2008). The same structure

applies wherever agents choose cases or exposure and are judged on raw outcomes: environmental-service contracts in which payments follow post-intervention observations, expert forecasting scored without conditioning on task difficulty (Murphy, 1993), and performance contracts with hidden counterfactuals. For empirical evaluation, the framework suggests estimating prospective natural hazards, excluding event-day information from timing rules, modeling selection into observed events, and distinguishing apparent outcomes from causal effects. These implications are inferences from a calibrated benchmark; we do not claim the safeguards were empirically tested.

5.5. Limitations

First, the eight locations are point-based stress tests. They span major rainfall regimes but do not establish global representativeness and are not matched to historical ritual communities. Spatially distributed precipitation, forecast information, and measurement error could alter hazards.

Second, the WTP, prices, costs, provider count, acceptance function, and observer rules are calibrated scenarios. They are not estimates of historical beliefs or payment schedules. Historical sources support the institutional relevance of rainmaking but not a universal claim of drought-contingent fee escalation (Espín-Sánchez & Gil-Guirado, 2022; Akong'a, 1987; Semenya, 2013).

Third, the hazard estimator conditions on a limited state vector. A provider could use richer forecasts, cloud observations, or local environmental knowledge. Such information would likely strengthen selection possibilities, but its effect is not estimated here.

Fourth, apparent success is defined by fixed rainfall thresholds and windows. The robustness grid demonstrates sensitivity expected from that definition, not a uniquely correct cultural attribution rule.

Fifth, pseudo-event rules can select identical or nearby dates, observer states persist over time, and attribution windows overlap. Two-way clustered covariance addresses dependence within region-years and region-rule series, but the scenario-WTP regressions remain exploratory and no multiplicity-adjusted confirmatory test is claimed.

Sixth, social utility is separated conceptually but not empirically measured. The analysis rejects the logical equivalence between meteorological and social value; it does not demonstrate that every ritual creates positive welfare.

Finally, a zero-effect design is intentionally a limiting case. It establishes what demand, selection, attribution, and learning can generate without efficacy. It cannot show that real-world environmental interventions are ineffective.

6. CONCLUSION

When providers choose their exposure, raw outcome rates combine baseline risk, selection, and causal performance. Under an exact zero-effect null, the causal term is removed and the remaining machinery still produces apparent performance: natural rainfall generates apparent success, drought-driven demand determines when communities seek action, selective acceptance shifts observed risk composition, attribution turns subsequent rain into perceived performance, and learning turns perceived performance into persistent reputation and price premia.

The positive contemporaneous price-success proposition is not supported: reputation-mediated mean prices and provider-price heterogeneity coexist with negative mean event-level price-success differences, because reputation is a stock of past attributed outcomes while current performance is a hazard-driven flow. For credible performance evaluation under heterogeneous environmental risk, the findings motivate prospective risk adjustment, disclosure of the case mix

that was not accepted, and separation of natural performance, causal performance, scenario price, transfers, costs, and plural values.

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Declarations

Data and code availability

Code, legally redistributable inputs, frozen analytical outputs, and manuscript-generation files are publicly available at https://github.com/bougtoir/pricing-the-rain. The repository preserves raw rainfall snapshots with machine-readable acquisition ledgers and checksums. Sources whose redistribution rights are unclear are not redistributed; the ledgers provide URLs, checksums, status notes, and retrieval instructions instead. Fictional pseudo-events and all simulation outputs are reproducible from the archived rainfall snapshots.

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Competing interests

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Author contributions

[AUTHOR INPUT REQUIRED: provide CRediT roles for each author.]

Ethics

The study uses public environmental data, bibliographic metadata, and simulated fictional events. It includes no human participants, personal data, or animal research.

Generative AI disclosure

During preparation of this work, the authors used Devin (Cognition AI) to assist with code development, manuscript drafting, and language organization. [AUTHOR CONFIRMATION REQUIRED: after using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.] No generative-AI image tool was used to create or alter the scientific figures, and no generative AI was used to fabricate data or references.

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